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"AI AGENT FORMULATION AND ANALYSIS"

Selected Agent: automated vacuum cleaner

• Problem in the Agent Context:

The main problem is to keep a multi-room house floor clean. The agent needs to navigate around randomly placed furniture, avoid falling down stairs, clean up dust or debris, and return to its charging dock before running out of battery.

• Problem Formulation :

1) Textual Formulation.

- States:-

The agent's current location and the status of that location.

- Initial State:-

The vacuum is at the charging dock, and all rooms have an unknown amount of dirt.

- Actions:-

Move left, Move Right, Move Up, Move Down.
Suck Dirt, Go to charger.

- Transition Model:-

Moving changes the agent's position;
sucking dirt changes the room status from Dirty to clean.

- Goal Test:-

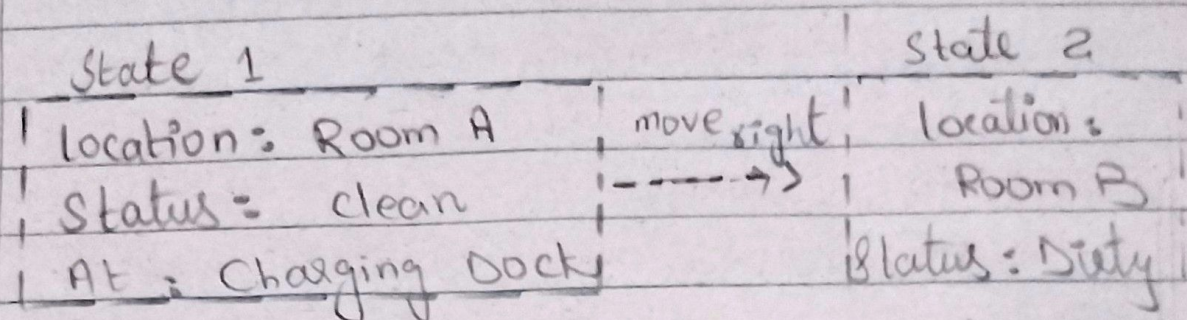
All reachable floor areas are completely clean.

- Path Cost:-

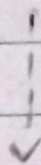
Every move costs 1 point of energy,
and sucking dirt costs 2 points.

The goal is to maximize cleanliness
while using the least energy.

• Graphical Formulation:-



(Suck Dirt)



[Goal State]

location: Room B

Status: Clean

Peas Evaluation:-

1) Performance Measure:

Percentage of dirt cleaned, total
Percentage time taken, & battery efficiency.

2) Environment:

↳ wooden floors, carpets, furniture,
dust & stairs.

- **Actuators:**
 - ↳ wheels for movement, spinning brushes, & a vacuum suction motor.

- **Sensors:**
 - ↳ Bumper sensors (to detect walls), infrared cliff sensors (to avoid stairs), and a dist detection sensor.

• **Environment Properties** ↗

- Partially observable -
- Stochastic
- Sequential
- Dynamic
- Continuous.
- Single Agent.